JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Basic Science & Humanities Department
2023-24 (Odd Semester)
MID SEMESTER EXAMINATION

Education to Eternity

॥ज्ञानम् सर्वार्थ साधनम्॥

Course: B. Tech.

Subject: Advanced Physics **Subject Code:** CS1T002/AI1T002/DS1T002 **Max Marks: 20**

Year/Semester: 1st Semester (First Year) **Date: 25/10/2023** **Duration:- 1 Hr**

Instructions to the Students:

1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

CO1: Describe the concept of LASER, optical fiber, types of semiconductors, PN junction diode characteristics, wave optics, electron Ballistics, quantum mechanics, various crystal structure parameters & X-rays.

CO2: Elaborate the types of LASER, optical fiber, Semiconductors, crystal structure, formation of Newton's ring, fringes in wedge shape thin film, effect of electric and magnetic field on motion of charge particle and significance of quantum mechanics.

CO3: Apply the concept of three and four level in LASER production, TIR in Optical fiber, classify the type of material based on current conduction, Bragg's law and X-ray diffraction, Interference for advanced application, illustrate the wave particle dualism of matter waves, motion and charged particle in E and B.

CO4: Analyze the behaviour of PN junction diode in FB and RB, compare the different types of LASER and optical fiber, correlate the motion of charged particles in uniform electric and magnetic fields for Bainbridge Mass spectrograph, the formation of fringes in thin film, behaviour of wave function and the types of crystal.

CO5: Justify physical significance of wave function, HUP, Schrodinger's wave equations, application of Hall effect, LASER & Optical Fibre, Wave Optics, Electron Ballistics and interpret he various crystal structure.

CO6: Design devices by using the concept of Laser, optical fibre, Electron ballistics, Semiconductor, crystals structure, wave optics and quantum mechanics.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	How LASER light differ from ordinary light?	1/CO1	1
Q.2	What is the principal on which optical signal propagate through optical fibre? TIR	1/CO1	1
Q.3	What type of Impurities added into intrinsic semiconductor to form N-type of Semiconductor? Pentavalent	1/CO1	1
Q.4	Give your justification to show that Semiconductor behave like insulator at 0° K.	2/CO2	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

Q.5	Compare the process of Spontaneous and Stimulated emission.	2/CO2	2
Q.6	Calculate Acceptance angle and Numerical Aperture for an Optical fibre having core refractive index 1.5 and cladding refractive index 1.40.	2/CO2	2
Q.7	Draw band energy diagram to show the variation of Fermi Energy with respect to temperature in N-type semiconductor.	2/CO2	2
Q.8	Classify the conductor, semiconductor and Insulator on the basis of band theory..	2/CO2	2

SECTION C

1. **Solve any two questions.**

Q.9	Explain the construction and working of He-Ne LASER with energy level diagram.	2/CO2	4
Q.10	Derive an expression for an Acceptance angle for the signal propagating through an Optical fibre.	3/CO3	4
Q.11	What is Fermi energy? Show that Fermi energy lies in the Middle of Band gap of Intrinsic Semiconductor.	4/CO4	4

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